

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

MONKEY BANANA PROBLEM

Aim:

To implement the Monkey Banana Problem using state-space representation and logical reasoning in Prolog.

Objective:

To find a sequence of actions through which the monkey can obtain the banana.

Algorithm:

- 1. Start.**
- 2. Represent the initial state of the monkey.**
- 3. Check whether the monkey is under the banana.**
- 4. If not, the monkey moves toward the box.**
- 5. The monkey pushes the box under the banana.**
- 6. The monkey climbs onto the box.**
- 7. The monkey gets the banana.**
- 8. Stop.**

Flowchart:

START

| v

Monkey at Door

| v

Move to Box

| v

Push Box Under Banana

| v

Climb Box

| v

Get Banana

| v

STOP

Prolog Program:

% Monkey Banana Problem

move(state(middle, onfloor, middle, hasnot), state(middle, onbox, middle, has)) :-

write('Monkey climbs onto the box and gets the banana. '), nl.

**move(state(Pos, onfloor, Pos, hasnot), state(Pos, onbox, Pos, hasnot)) :-write('Monkey
climbs onto the box. '), nl.**

move(state(Pos1, onfloor, Box, hasnot), state(Pos2, onfloor, Box, hasnot)) :-Pos1 \= Pos2,

write('Monkey moves from '), write(Pos1),

write(' to '), write(Pos2), nl.

**move(state(Pos, onfloor, Pos, hasnot), state(Pos, onfloor, Pos, hasnot)) :-write('Monkey
is under the box. '), nl.**

**monkey_gets_banana :-write('Monkey is at the middle. '), nl, write('Monkey climbs the
box. '), nl, write('Monkey gets the banana. '), nl, write('Banana obtained!'), nl.**

Query

?- monkey_gets_banana.

Output

Result:

Thus, the Monkey Banana Problem is successfully represented using Prolog.